

to the transmission axis of the detector. Intensity detected becomes $I \cos^2 90$

8(a) Binding energy is the minimum amount of energy required to separate a nucleus into its constituent nucleons such that they are infinitely far apart. B2

Or

Binding energy is the energy released when the nucleus forms from its separate protons and neutrons.

Comments:

Energy is required to separate a nucleus into its constituent nucleons while energy is released when the nucleus is formed from its separate protons and neutrons.

(b)(i) Mass defect $\Delta m = 88m_p + (226 - 88)m_n - M_{Ra}$
 $= 88(1.00783u) + 138(1.00867u) - 226.025u$
 $= 1.8605u$

M1

Binding energy per nucleon $= \frac{\Delta mc^2}{226}$

$= \frac{1.8605(1.66 \times 10^{-27})(3.00 \times 10^8)^2}{226}$

M1

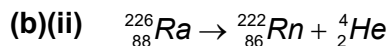
$= 1.2299 \times 10^{-12} J$

$= 7.68 MeV$

A1

Comments:

226.025u is the mass of $^{226}_{88}Ra$ nucleus. Mass defect of $^{226}_{88}Ra$ nucleus is the difference between the mass of the nucleus and the sum of masses of its individual nucleons.



Energy released

$= m_{reactants}c^2 - m_{products}c^2$

$= (226.025 - 222.018 - 4.00260)(1.66 \times 10^{-27})(3.00 \times 10^8)^2$

M1

$= 6.5736 \times 10^{-13} J$

$= 4.11 MeV$

A1

(c)(i) Random – do not know when or which particular nucleus will decay. But there is a constant probability of decay (decay constant) B1

B1

Spontaneous – not affected by external factors.

Comments:

This part is generally quite badly done. Most of them were not able to define the 2 terms correctly.

(c)(ii) The half-life of $^{222}_{86}Rn$ is very short (only 3.8 days). It decays very quickly. But $^{222}_{86}Rn$ is the product of other radioactive decay (which probably has a long half-life) and so takes a long time to be produced. Thus they can still be found on Earth.

B1

B1

Comments:

This part is generally quite badly done. Most of the students were not able to explain why it is rare.

(c)(ii) $A = \lambda N$

$N = \frac{200}{2.1 \times 10^{-6}}$

$= 9.52 \times 10^7$

M1

$N = \frac{m}{M_r} N_A$

$$\text{Mass} = \frac{9.52 \times 10^7}{6.023 \times 10^{23}} \times 222 \times 10^{-3}$$

$$= 3.5 \times 10^{-17} \text{ kg}$$

M1

A1

Or

$$\text{Mass} = 9.52 \times 10^7 \times 222 \times 1.66 \times 10^{-27}$$

$$= 3.5 \times 10^{-17} \text{ kg}$$

Comments:

This part is quite well done. The only problem is that some of the students did not convert the unit of grams to kilograms, thus losing 1 mark.

(d)(i) $A = A_0 e^{-\lambda t}$

$$\frac{8.5}{100} A_0 = A_0 e^{-\frac{\ln 2}{t_1} t}$$

M1

$$\ln\left(\frac{8.5}{100}\right) = -\left(\frac{\ln 2}{5700}\right) t$$

$$t = 20300 \text{ years}$$

A1

Comments:

This part was well attempted.

(d)(ii) Probability of decay per unit time

$$= \lambda$$

$$= \frac{\ln 2}{t_1} = \frac{\ln 2}{5700(365)(24)(60)(60)} = 3.86 \times 10^{-12} \text{ s}^{-1}$$

A1

Comments:

Some students do not realise that the probability of decay is the decay constant. And some students forgot to convert year to second. Students are advised to look at the unit requirement shown at the answer to be aware.

(d)(iii) Carbon-14 cannot be used to determine the age of the stone.

Carbon-14 can only be used for living organism. Once it dies, it ceases to acquire ^{14}C (from atmosphere or diet), but the ^{14}C within its biological material at that time will continue to decay, and so the ratio of ^{14}C to ^{12}C in its remains will gradually decrease. The change in ratio can be used for carbon dating. A stone does not acquire ^{14}C .

B1

B1

Comments:

This part requires understanding of carbon dating which majority of the candidates who attempted this question do not have. Hence it was not well answered.

(d)(iv) In beta decay, a neutron changes into a proton and electron. The electron is emitted as beta particle. The proton remains in the nucleus. Hence, the neutron number decreases by 1 and the proton number increases by 1.

M1

Comments:

The question asks for "state and explain" hence merely stating n -1 and p +1 is insufficient as reason has to be given.